



ETHI 401

AI Biometric Voter Verification for Peru

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A multimodal AI-powered voter identity verification system for Peru using three biometric methods:



Fingerprint Matching



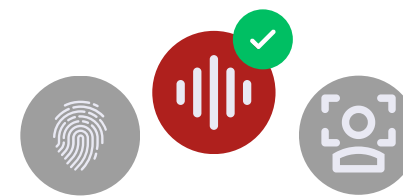
Facial Recognition



Voice Recognition

Introduction: What are we building?

Authentication Model (two levels of redundancy):



Any one method works



Each method runs 3
times — 2/3 attempts
must pass



If all methods fail →
Manual assistance



Reference: Similar to Estonia's e-ID system [1] and India's Aadhaar [2]



Introduction: System Design

Key design principle: Biometrics stored only on a secure chip embedded in the voter's ID card (no central database, no internet required, offline capable)





Potential Harms

AI Principles: Privacy, Reliability, Transparency



Security risks

- Voice synthesizable with widely available tools [3]
- Fingerprint residue can be harvested and duplicated
- Facial recognition defeatable with photographs
- 3 biometric types = breach is 3x more damaging [4]
- More hardware = greater failure risk in rural stations [5][6]



Potential Harms

AI Principles: Privacy, Reliability, Transparency

Reliability risks

- Voice degrades in noisy environments [7]
- Fingerprint affected by humidity or extreme dryness [8]
- Facial recognition fails on tanned/weathered skin
- System must explain why verification failed (without this, voter trust collapses)



Potential Harms: Who is most at risk?

AI Principles: Fairness and Non-Discrimination

Group	Risks
Indigenous/mestizo majority	Underrepresented in most AI training data
Quechua, Aymara, Amazonian speakers	Disadvantaged in voice models vs. Spanish speakers [7]
Older manual laborers	Fingerprints worn down over lifetime of work [9]
Women and darker skin tones	Facial recognition underperforms significantly [8][10]
Rural and remote populations	Limited electricity, connectivity, longer travel to polling stations [11]

Core concern: Peru's most vulnerable voters are the most likely to be failed by the system — demands a robust non-biometric fallback [12]



Risk Mitigation

AI Principles: Privacy by Design, Accountability

System design mitigations

- On-chip storage only: no central biometrics database [13][14]
- Offline capable: no cloud dependency [15]
- Cascading authentication: only 1 method needs to succeed
- Consent-first enrollment [16]
- Human fallback always available





Risk Mitigation

AI Principles: Privacy by Design, Accountability

Governance mitigations

- Locally representative training data [17]
- Demographic bias audits before each election
- Ongoing monitoring and feedback cycle post-election
- Regular model retraining as population patterns shift



Risk Mitigation

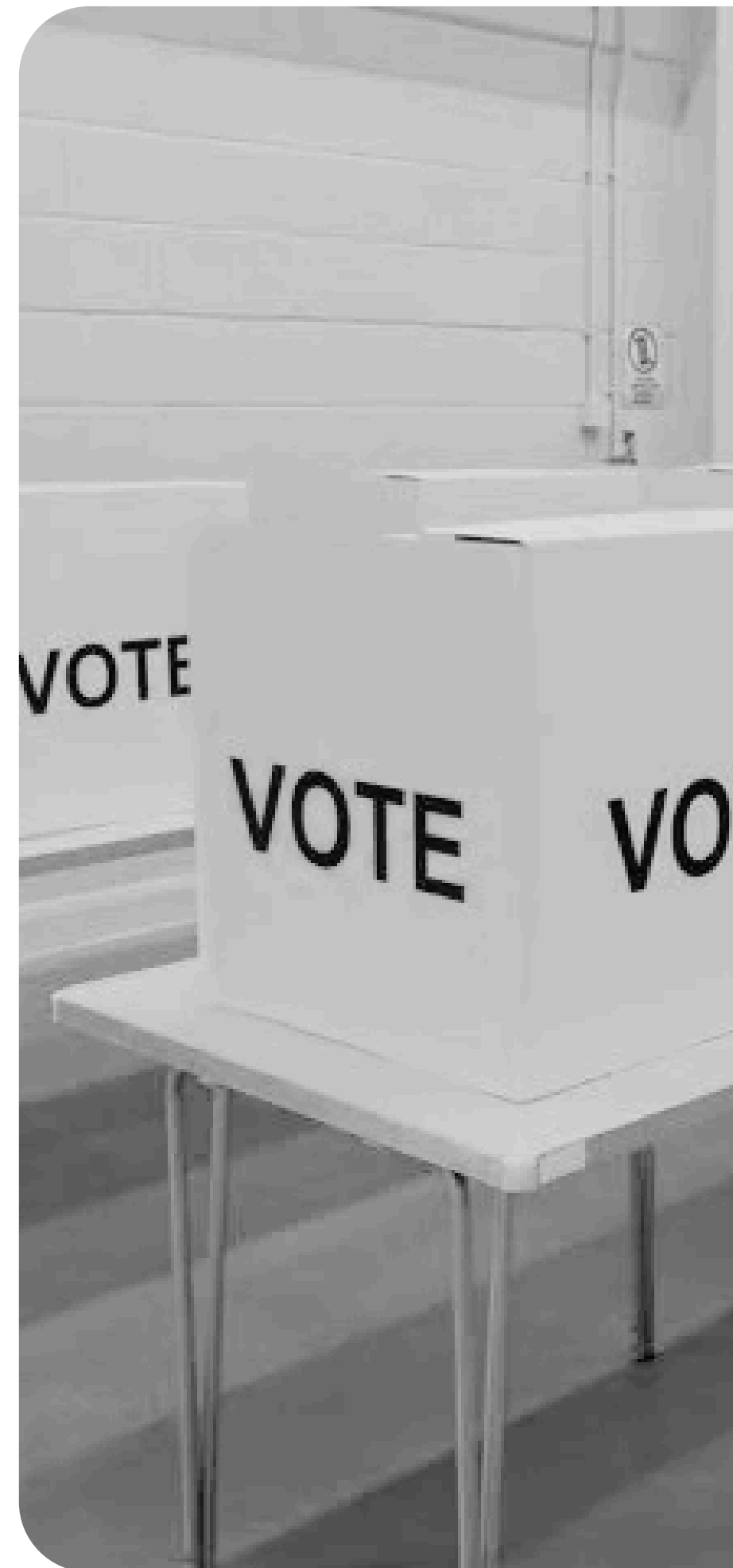
AI Principles: Robustness and Safety

Operational resilience

- Fully offline
- Human verification available if biometrics fail or consent not granted
- Offline chip design supports future mobile polling for remote voters

Maintenance plan

- Voluntary biometric updates through approved institutions
- Scheduled and random audits of models and capture hardware
- Mandatory post-election bias audit



Unavoidable Biases

AI Principles: Fairness (acknowledging what cannot be fully solved)

Biometric biases



Underperforms on women and darker skin tones [8] [10]



Struggles with worn, scarred, aged fingertips



Underperforms on minority languages; ambient noise [7]

Structural biases



20%+ of Peru's 34M live in remote areas — longer travel, lower turnout



Rural/Indigenous communities lack electricity and internet [11]





Future Plans

- Pre-deployment testing following NIST recommendations — especially for racial and gender bias in Peruvian demographics [17]
- Pilot in urban polling stations before rural rollout
- Long-term: mobile polling capability for remote voters using the existing offline chip design
- Keep honest commitment: some bias is structural and unavoidable. The response is transparency, fallback mechanisms, and continuous auditing



Conclusions

What we proposed:

On-chip multimodal biometric verification (fingerprint, face, and voice) extending Peru's RENIEC system, not replacing it.

What the analysis revealed:

- Greatest risks fall on Peru's most marginalized voters
- On-chip design, human fallback, and consent-first enrollment are the most critical mitigations
- Some biases are unavoidable — transparency and fallback are the appropriate response

Core governance principle:

Increasing election security must never come at the cost of disenfranchising the voters the system is meant to protect.

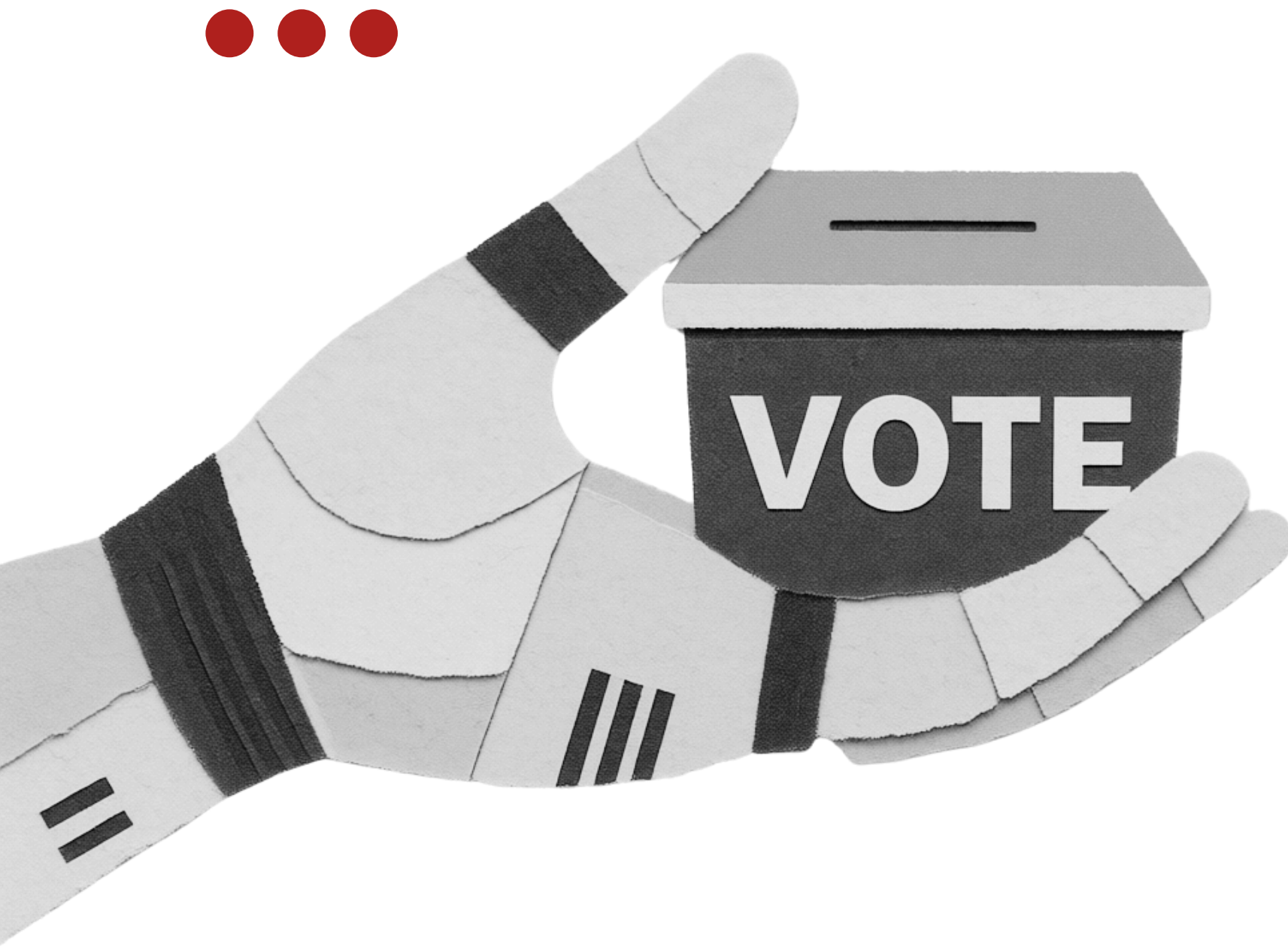




Sources

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Thank you!

Any questions?

AI Usage Statement: "AI was used to support brainstorming, structural organization, formatting, and refinement. All ideas, analysis, and final content decisions were reviewed and verified by the authors."